COURSE CODE/TITLE: CPE 010-CPE21S1 - Data Structures and Algorithms	**SECTION: CPE21S1**
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ASSIGNMENT NO. 6.1 Linear and Binary Search

1. PROCEDURE OUTPUT

Answer the following questions:

What is a search tree in data structures?

> A search tree is a tree-based data structure that organizes elements according to their values so that they can be searched efficiently. An example of a search tree is a Binary Search Tree (BST), where the values smaller than a node are placed in its left subtree, while larger values are placed in its right subtree.

What are the different types of search algorithms in data structures? Differentiate each type of search.

> The two common types of search are Linear Search and Binary Search. Linear Search checks each element one by one, starting from the first element, until the target is found or the list ends. It can be used whether the data is sorted or unsorted. Its time complexity is O(n). Binary Search, on the other hand, repeatedly divides a sorted list into two halves. It compares the target with the middle element and continues searching only in the appropriate half. Its time complexity is O(log n).

What operations / implementations can be performed using binary and linear search operations?

> The implementations that can be performed using binary search operations include the searching for a specific element or its index in a sorted array/vector, finding the start and end points of boundaries by repeatedly dividing the searching space in half, finding floor as well as ceiling values, and finding the insertion point of a sorted away. On the other hand, linear search operations allow us to find an element within an unsorted list, array, etc., looking through the dataset to find minimum and maximum values, seeing how many occurrences of a specific element happens or finding duplicates, and allows searching through data structures like linked lists where index-based access isn’t available.

What are the advantages in using binary search trees as data structure?

> Binary Search Trees or BST offers an efficient search time, and has an ordered structure that stores the elements in a sorted order, making it simple to locate the next and previous elements. Additionally, BST allows for efficient dynamic insertion and deletion, allowing elements to be added at any point smoothly without predefined size limits. Lastly, Its design allows it to function effectively as a doubly ended priority queue, where both the maximum and minimum values can be maintained efficiently.

Give an example program using binary search and Linear search.

> **Linear Search:**

Unsorted Shopping Lists or To-Do Apps: Checking off items on a daily checklist or searching an unordered list of notes scans line-by-line from top to bottom.

Small Database or Form Validation: Scanning a short list of 5–10 text fields (like checking if a newly entered form field matches a small list of restricted characters) uses sequential checking.

Basic Text Search in Small Documents: Simple text editors scanning a short, unsorted string or document for a specific character without prior indexing.

> **Binary Search:**

Search Engine Indexes (Google, Bing): When search engines process a query, they query massive, pre-sorted index tables using logarithmic divide-and-conquer searching to return results in milliseconds.

Database Indexing & SQL Queries: Relational databases (like PostgreSQL or MySQL) use indexed columns (B-Trees/Binary Search principles) to locate specific user IDs or record numbers instantly out of millions of rows.

Looking Up Words in a Physical Dictionary: Opening a dictionary to the middle, checking if your word comes before or after that page, and discarding half the pages at each step.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

N/A

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **None**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **None**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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